

AWS SQS, AWS MQ & AWS SECRETS MANAGER

AWS SQS Hands-on Implementation

Objective: Implement Amazon SQS by creating a queue, sending and receiving messages, configuring a Dead Letter Queue (DLQ), enabling long polling, and verifying queue operations.

Prerequisites

Before starting, ensure you have:

- AWS Account
- IAM User with SQS permissions (or AdministratorAccess)
- AWS Region selected (recommended: **US East (N. Virginia) - us-east-1**)

Step 1: Login to AWS Console

1. Open your AWS Management Console.
2. Select the region **US East (N. Virginia)**.
3. In the search bar, type **SQS**.
4. Click **Simple Queue Service (SQS)**.

Step 2: Create a Standard Queue

1. Click **Create queue**.
2. Under **Queue type**, select **Standard**.
3. Enter the queue name: **employee-standard-queue**

The screenshot shows the AWS Management Console interface for creating a new Amazon SQS queue. At the top, there's a navigation bar with the AWS logo, a search bar, and the user's account information (Rajkumari). The main header shows the breadcrumb 'Amazon SQS > Queues > Create queue'. A blue notification banner at the top states: 'SQS fair queues is now available for standard queues. Automatically manage noisy neighbors in your queues with fair queues, a new feature that limits noisy neighbor impact across all message groups. Add a group identifier to your messages, and SQS re-orders messages to ensure no single tenant impact the time in queue for any other tenants.' Below this, the 'Create queue' section is divided into 'Details' and 'Configuration' tabs. The 'Details' tab is active, showing two radio button options for queue type: 'Standard' (selected) and 'FIFO'. The 'Standard' option is described as 'At-least-once delivery, message ordering isn't preserved' with sub-points 'At-least once delivery' and 'Best-effort ordering'. The 'FIFO' option is described as 'First-in-first-out delivery, message ordering is preserved' with sub-points 'First-in-first-out delivery' and 'Exactly-once processing'. A warning message states: 'You can't change the queue type after you create a queue.' Below the queue type options, there's a 'Name' field with the text 'employee-standard-queue' and a note: 'A queue name is case-sensitive and can have up to 80 characters. You can use alphanumeric characters, hyphens (-), and underscores (_).' The 'Configuration' tab is partially visible at the bottom.

Configure Queue Settings

Setting	Value
Queue Type	Standard
Visibility Timeout	30 Seconds
Message Retention	4 Days
Delivery Delay	0 Seconds
Maximum Message Size	256 KB (default)
Receive Message Wait Time	0 Seconds (we'll change this later)
Leave all other settings at their default values.	
Click Create Queue .	

Configuration [Info](#)

Set the maximum message size, visibility to other consumers, and message retention.

Visibility timeout [Info](#)

30 Seconds

Should be between 0 seconds and 12 hours.

Message retention period [Info](#)

4 Days

Should be between 1 minute and 14 days.

Delivery delay [Info](#)

0 Seconds

Should be between 0 seconds and 15 minutes.

Maximum message size [Info](#)

1024 KIB

Should be between 1 KIB and 1024 KIB.

Receive message wait time [Info](#)

0 Seconds

Should be between 0 and 20 seconds.

Encryption [Info](#)

Amazon SQS provides in-transit encryption by default. To add at-rest encryption to your queue, enable server-side encryption.

Server-side encryption

☐ Disabled

☒ Enabled

Encryption key type

☒ Amazon SQS key (SSE-SQS)

An encryption key that Amazon SQS creates, manages, and uses for you.

Step 3: Explore Queue Details

Open the newly created queue and observe:

- Queue URL
- ARN
- Queue Type
- Encryption (SSE-SQS by default)
- Access Policy
- Metrics

Amazon SQS [Queues](#) > employee-standard-queue

Queue employee-standard-queue created successfully
You can now send and receive messages.

Notifications 0 0 0 1 1 0 0

employee-standard-queue [Edit](#) [Delete](#) [Purge](#) [Send and receive messages](#) [Start DLQ redrive](#)

Details [Info](#)

Name employee-standard-queue	Type Standard	ARN arn:aws:sqs:us-east-1:57183381790:employee-standard-queue
Encryption Amazon SQS key (SSE-SQS)	URL https://sqs.us-east-1.amazonaws.com/57183381790/employee-standard-queue	Dead-letter queue -

[More](#)

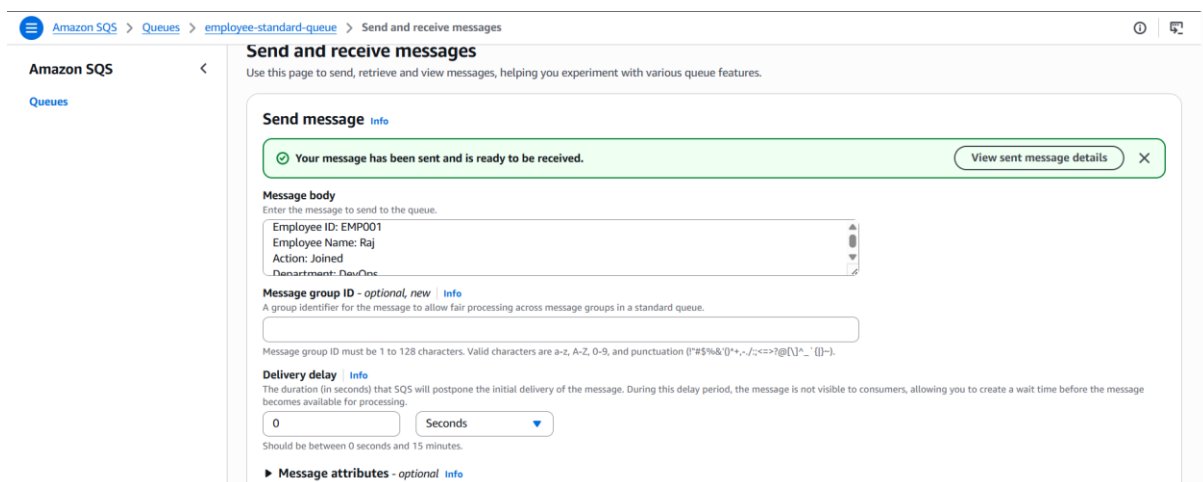
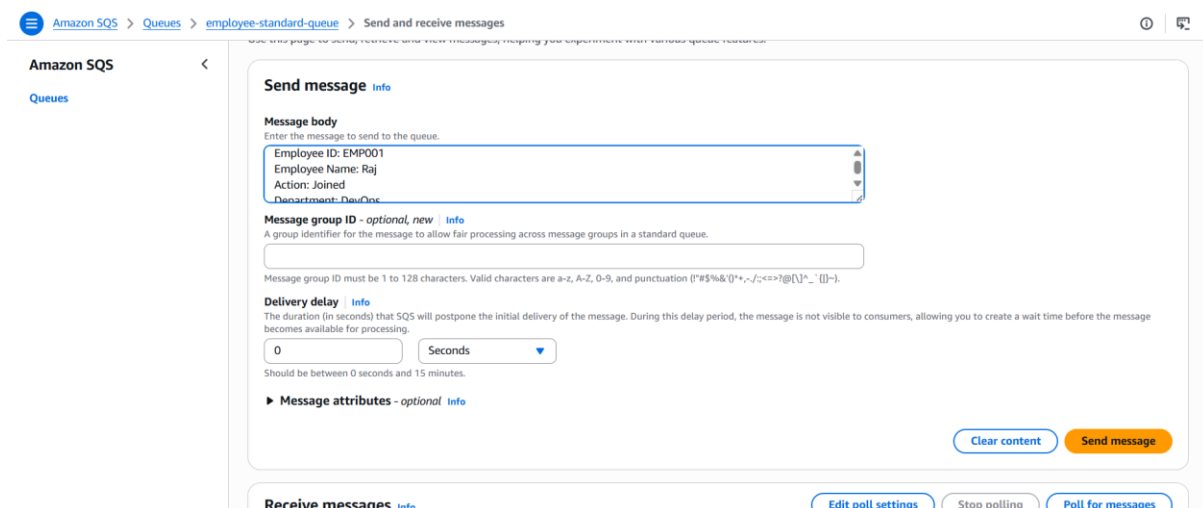
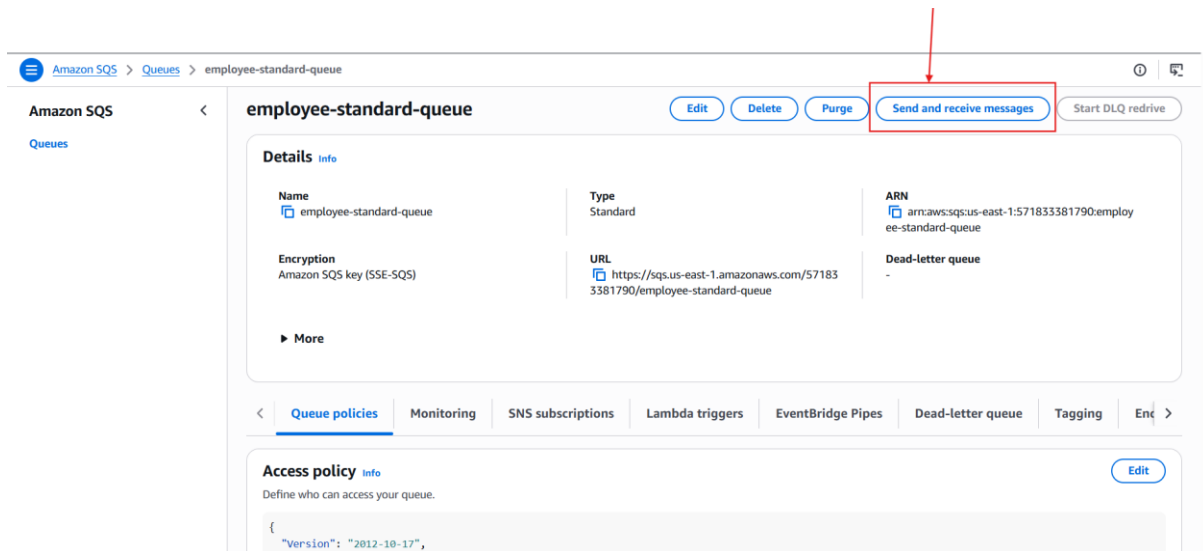
[Queue policies](#) [Monitoring](#) [SNS subscriptions](#) [Lambda triggers](#) [EventBridge Pipes](#) [Dead-letter queue](#) [Tagging](#) [Encryption](#)

Access policy [Info](#) [Edit](#)

Step 4: Send Messages

1. Open the queue.

2. Click **Send and receive messages**.
3. Under **Message body**, send the following messages one by one.



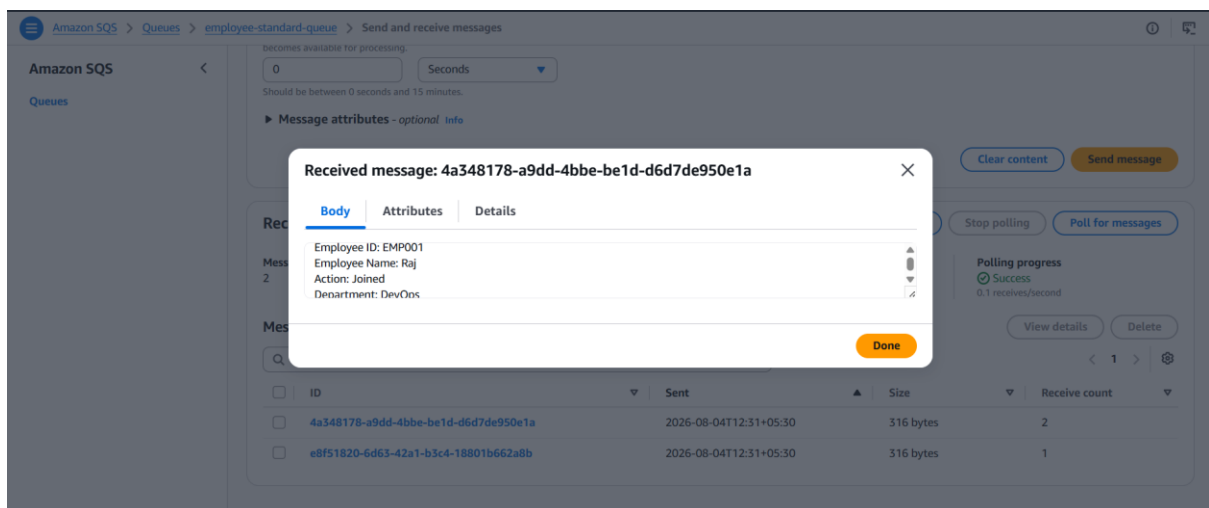
Step 5: Receive Messages

1. Stay on the **Send and receive messages** page.
2. Scroll to **Receive messages**.
3. Click **Poll for messages**.

You should see all four messages.

Observe:

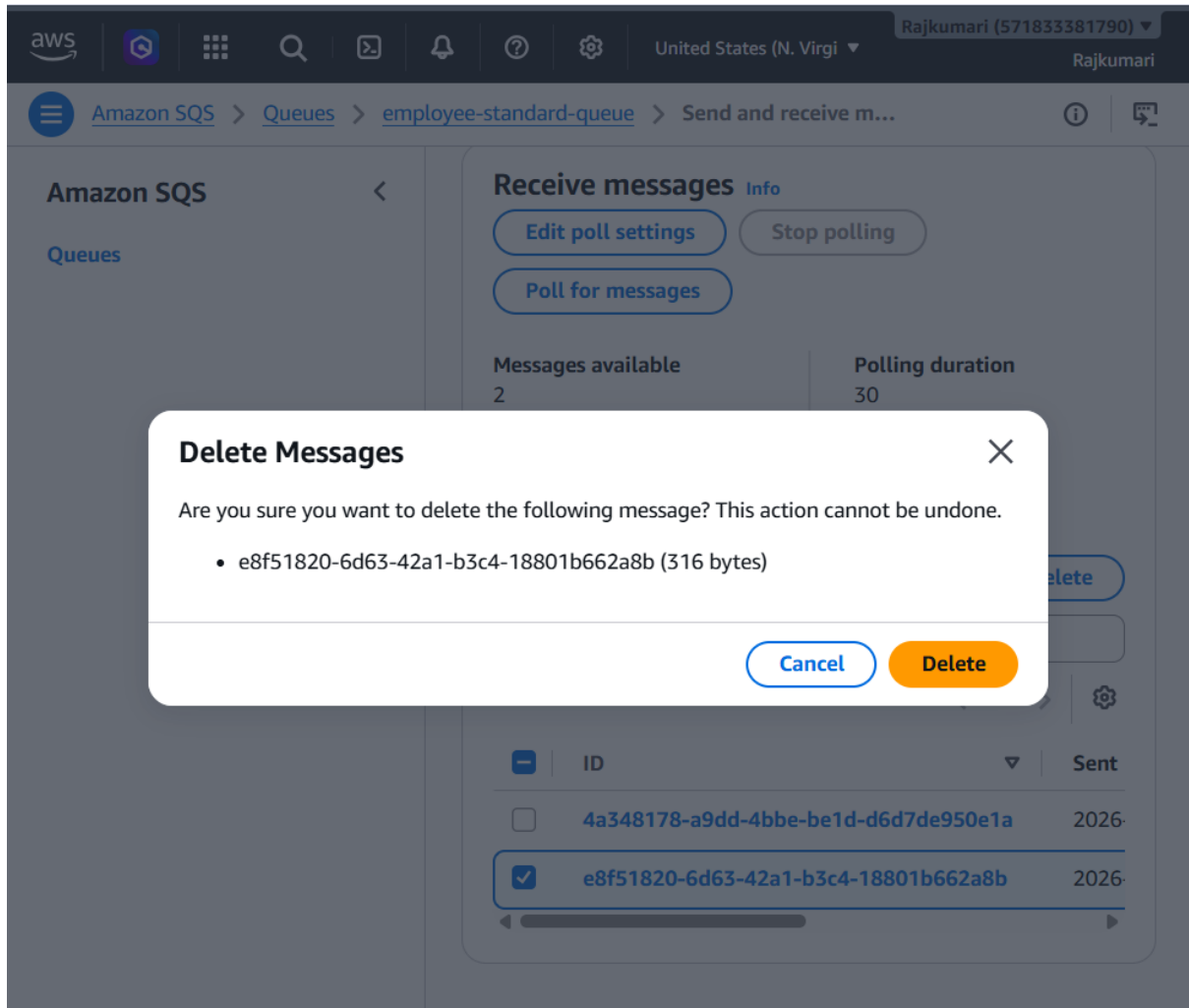
- Message ID
- Sent Timestamp
- Approximate Receive Count
- Receipt Handle



Step 6: Delete Messages

1. Select one of the received messages.
2. Click **Delete**.

Poll again to verify the remaining messages.

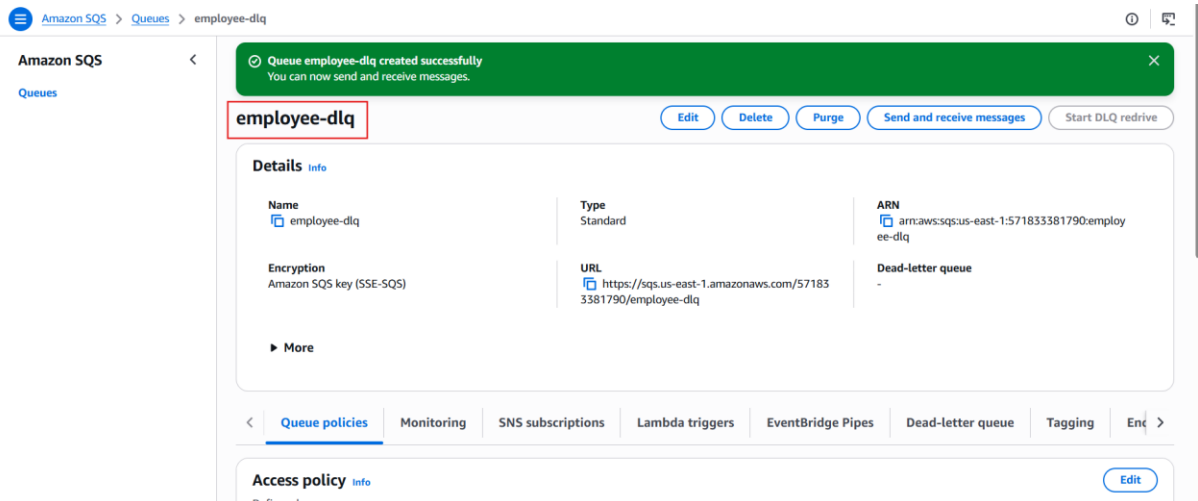


Step 7: Create a Dead Letter Queue (DLQ)

1. Return to the SQS dashboard.
2. Click **Create queue**.
3. Choose **Standard**.
4. Name the queue:

employee-dlq

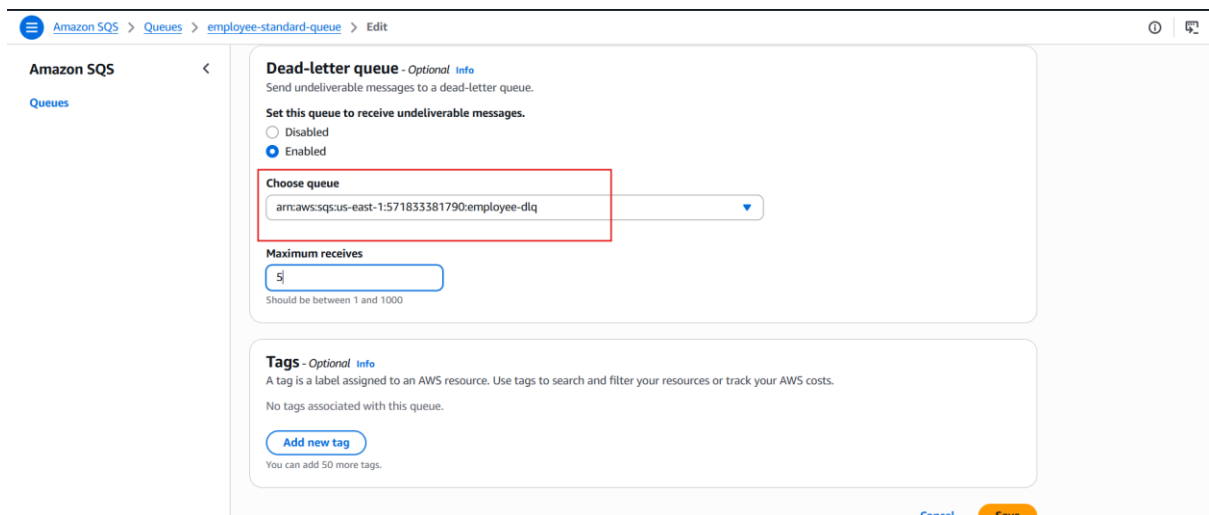
Leave other settings as default and click **Create Queue**.

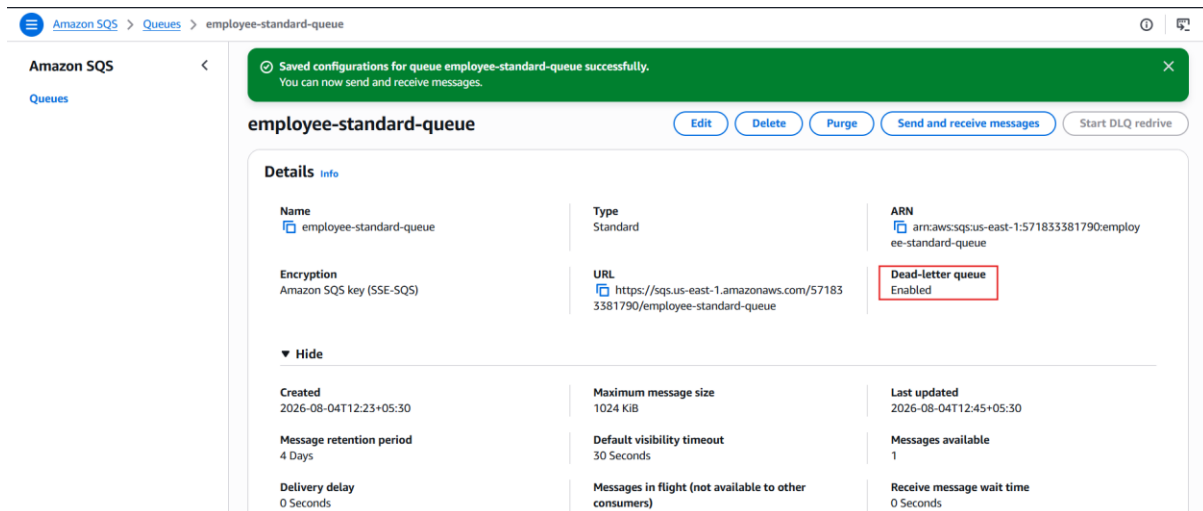


Step 8: Attach the DLQ to the Main Queue

1. Open **employee-standard-queue**.
2. Click **Edit**.
3. Go to the **Dead-letter queue** section.
4. Enable **Use a dead-letter queue**.
5. Select **employee-dlq**.
6. Set **Maximum receives** to: 5
7. Save changes.

What it means: If a message fails processing five times, SQS automatically moves it to the DLQ for later investigation instead of retrying indefinitely.

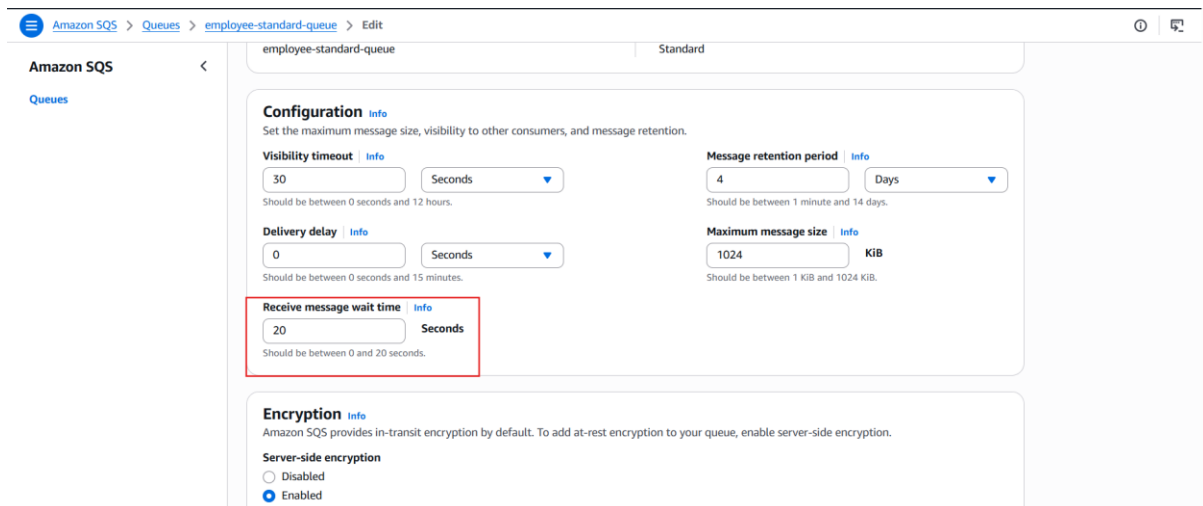




Step 9: Enable Long Polling

1. Open **employee-standard-queue**.
2. Click **Edit**.
3. Under **Receive message wait time**, set: 20 Seconds
4. Save the changes.

Why? Long polling waits for messages before responding, reducing empty responses and unnecessary API calls.

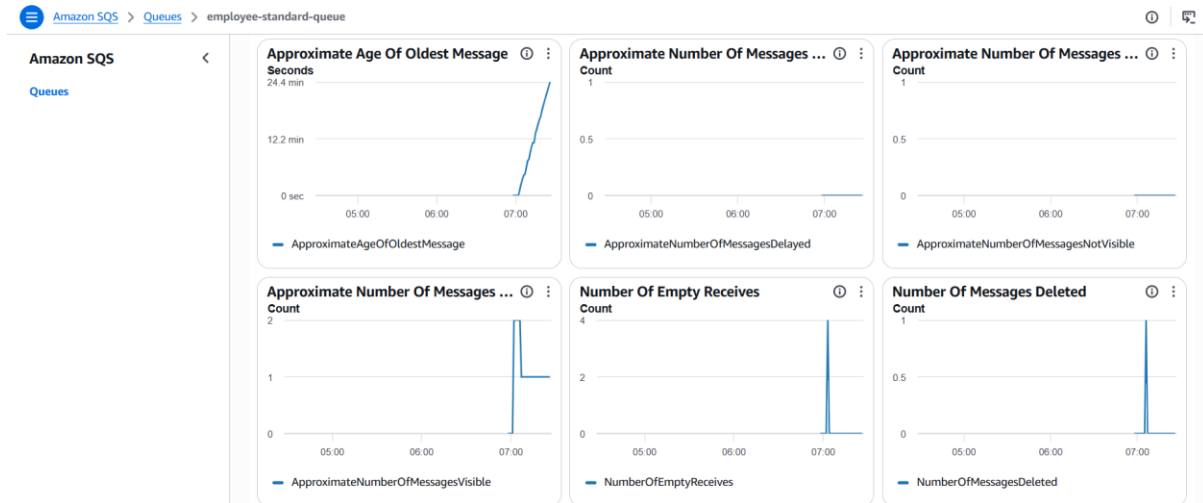


Step 10: Review Monitoring Metrics

Open the **Monitoring** tab and review:

- Number of messages sent
- Number of messages received

- Number of messages deleted
- Approximate number of visible messages
- Approximate age of oldest message



In addition to using the AWS Management Console, we can also create and manage AWS SQS services using the AWS CLI by using below Commands

Step 11: AWS CLI Practice

If you have the AWS CLI configured, run:

List queues:

```
</> Bash
aws sqs list-queues
```

Create a queue:

```
</> Bash
aws sqs create-queue \
  --queue-name employee-standard-queue-cli
```

Send a message:

```
</> Bash
aws sqs send-message \
  --queue-url <QUEUE_URL> \
  --message-body "Employee ID: EMP005, Action: Joined"
```

Receive messages:

```
</> Bash
aws sqs receive-message \
  --queue-url <QUEUE_URL>
```

Delete a message:

```
</> Bash
aws sqs delete-message \
  --queue-url <QUEUE_URL> \
  --receipt-handle <RECEIPT_HANDLE>
```

Step 12: Cleanup

To avoid unnecessary charges:

1. Delete employee-standard-queue.
2. Delete employee-dlq.

Amazon MQ Hands-on Implementation

NOTE: Amazon MQ is not included in the AWS Free Tier. Therefore, this document covers the step-by-step hands-on implementation and configuration process only. Creating an Amazon MQ broker is a billable operation and may incur AWS charges. To avoid unexpected costs, the broker creation step was not performed.

Objective: Deploy an Amazon MQ RabbitMQ broker, configure networking and authentication, verify its status, and understand how enterprise applications communicate using a managed message broker.

Prerequisites

Before creating the broker:

- AWS account
- IAM user with Amazon MQ permissions
- Region selected (for example, **US East (N. Virginia)**)
- A VPC with at least one subnet (Amazon MQ will be deployed into your VPC)

Step 1: Open Amazon MQ

1. Sign in to the AWS Management Console.
2. Select your AWS Region.
3. Search for **Amazon MQ**.
4. Open the Amazon MQ service.

Step 2: Create a Broker

1. Click **Create broker**.

Step 3: Select Broker Engine

Choose:

- **RabbitMQ**

RabbitMQ is widely used in microservices, event-driven applications, and DevOps environments.

Click **Next**.

The screenshot shows the 'Select broker engine' step in the Amazon MQ console. On the left, a vertical progress bar lists five steps: Step 1 (Select broker engine, highlighted with a blue circle), Step 2 (Select deployment mode), Step 3 (Configure settings), Step 4 (Network settings), and Step 5 (Review and create). The main content area is titled 'Select broker engine' with an 'Info' link. Below the title, a box labeled 'Broker engine types' contains the instruction 'Select broker engine' and a sub-note: 'A broker engine is a type of message broker that runs on Amazon MQ.' Two radio button options are presented: 'RabbitMQ' (selected with a blue dot) and 'Apache ActiveMQ' (unselected). At the bottom right of the form, there are 'Cancel' and 'Next' buttons.

Step 4: Configure Broker Details

Enter the following values:

Setting	Value
Broker Name	employee-rabbitmq-broker
Deployment Mode	Single-instance broker
Broker Instance Type	mq.t3.medium(<i>or the smallest available option</i>)
Engine Version	Default latest version

Step 5: Create Administrator Credentials

Create the broker administrator account.

Example:

Field Value

Username raji

Password *****

The screenshot shows the 'Select deployment mode' step in the Amazon MQ console. On the left, a progress bar indicates the steps: Step 1 (Select broker engine), Step 2 (Select deployment mode - active), Step 3 (Configure settings), Step 4 (Network settings), and Step 5 (Review and create). The main content area is titled 'Select deployment mode' with an 'info' link. It contains two options: 'Single-instance broker' (selected) and 'Cluster deployment'. The 'Single-instance broker' option is described as 'One broker in one Availability Zone. Use for development and testing, or for smaller workloads.' and is accompanied by a diagram of a single node. The 'Cluster deployment' option is described as 'Three single-instance brokers connected in a mesh network across multiple Availability Zones. Use for production workloads that require high availability and message durability.' and is accompanied by a diagram of three nodes in a mesh. At the bottom right, there are 'Cancel', 'Previous', and 'Next' buttons.

Step 6: Configure Network

Amazon MQ must be deployed inside a VPC.

Select:

Setting Example

VPC Your existing VPC

Subnet One private subnet (recommended)

Security Group Existing security group or create a new one

The screenshot shows the 'Network settings' step in the Amazon MQ console. On the left, a progress bar indicates the steps: Step 1 (Select broker engine), Step 2 (Select deployment mode), Step 3 (Configure settings), Step 4 (Network settings - active), and Step 5 (Review and create). The main content area is titled 'Network settings' with an 'info' link. It contains a section 'Broker network access type' with two options: 'Public' (selected) and 'Private'. The 'Public' option is described as 'A broker that can be accessed directly from the internet, or outside of a VPC.' and is accompanied by a diagram of a single node. The 'Private' option is described as 'A broker instance that is not accessible over the internet and can only be accessed within a VPC or through AWS Resource Access Manager (RAM).' and is accompanied by a diagram of three nodes in a mesh. Below these options, there is a note: 'The broker is assigned public IP addresses at the time of broker creation and can be accessed directly from the internet via a public DNS name and IP address, outside of a VPC.' At the bottom right, there are 'Cancel', 'Previous', and 'Next' buttons.

The screenshot shows the 'Configure settings' page in the Amazon MQ console. On the left, a vertical navigation pane lists five steps: Step 1: Select broker engine, Step 2: Select deployment mode, Step 3: Configure settings (highlighted with a blue circle), Step 4: Network settings, and Step 5: Review and create. The main content area is titled 'Configure settings' and contains several configuration fields:

- Broker name:** A text input field containing 'employee-rabbitmq-broker'. Below it, a note states: 'Must be 1-50 characters long. Limited to alphanumeric characters, dashes, and underscores.'
- Broker engine version:** A dropdown menu showing '4.2'. Below it, a note states: 'For brokers on this version, Amazon MQ will always manage upgrades to the latest supported patch version.'
- Turn on automatic minor version upgrades:** A checkbox that is checked. Below it, a note states: 'When you turn on automatic minor version upgrades, Amazon MQ upgrades your broker to new patch versions as they become available.'
- Broker instance type:** A dropdown menu showing 'mq.m7g.medium'. Below it, a note states: 'Use for regular development, testing, and production workloads. 1 vCPU 4Gb RAM High Network'.
- Deployment mode:** A text field showing 'Single-instance broker'.
- Estimated deployment time:** A text field showing '20 minutes'.
- Storage type:** A text field showing 'Amazon Elastic Block Store'.
- Broker engine:** A text field showing 'RabbitMQ'.

Step 7: Review and Create

1. Review all settings.
2. Click **Create broker**.

The broker status will initially show:

Creation in progress

The screenshot shows the 'Review and create' page in the Amazon MQ console. On the left, the same vertical navigation pane is shown, with Step 5: Review and create highlighted with a blue circle. The main content area is titled 'Review and create' and contains three sections, each with an 'Edit' button in the top right corner:

- Step 1: Broker engine**
 - Engine type:** A text field showing 'Broker engine RabbitMQ 4.2'. Below it, a note states: 'For brokers on this version, Amazon MQ will always manage upgrades to the latest supported patch version.'
- Step 2: Deployment mode**
 - Deployment mode options:** A text field showing 'Deployment mode Single-instance broker'.
- Step 3: Configuration**
 - Details:** A text field showing 'Details'.

Step 8: Verify Broker Status

Once provisioning completes, the broker status should change to:

Running

Review:

- Broker ARN

- Engine (RabbitMQ)
- Engine version
- Deployment mode
- Endpoints

Step 9: Explore the Broker

Open the broker and review:

- Overview
- Configuration
- Networking
- Logs
- Monitoring

Document what each section contains.

Step 10: Reboot the Broker

1. Select the broker.
2. Choose **Reboot**.
3. Confirm the action.
4. Wait for the status to return to **Running**.

Step 11: Delete the Broker (Cleanup)

After completing the lab:

1. Select the broker.
2. Choose **Delete**.
3. Confirm the deletion.

AWS Secrets Manager Hands-on Implementation

Note: AWS Secrets Manager is not included in the AWS Free Tier. Therefore, this document covers the step-by-step hands-on implementation and configuration process only. Creating and storing a secret in AWS Secrets Manager is a billable operation and may incur AWS charges. To avoid unexpected costs, the secret creation step was not performed.

Objective: To securely create, store, retrieve, update, and delete secrets using AWS Secrets Manager without exposing sensitive information in application code.

Prerequisites

Before starting, ensure you have:

- AWS Account
- IAM User with Secrets Manager permissions
- AWS Region selected (recommended: **US East (N. Virginia)**)

Step 1: Open AWS Secrets Manager

1. Log in to the AWS Management Console.
2. Select your AWS Region.
3. Search for **Secrets Manager**.
4. Click **AWS Secrets Manager**.

Step 2: Create a Secret

1. Click **Store a new secret**.

Step 3: Choose Secret Type

You will see multiple options:

- Credentials for Amazon RDS database
- Credentials for Amazon Redshift
- Credentials for Amazon DocumentDB

- Other type of secret

Select: ☒ Other type of secret

Choose secret type

Secret type Info

☐ Credentials for Amazon RDS database
 ☐ Credentials for Amazon DocumentDB database
 ☐ Credentials for Amazon Redshift data warehouse
 ☒ Other type of secret
API key, OAuth token, other.

Key/value pairs Info

Key/value Plaintext

[+ Add row](#)

Step 4: Enter Secret Values

Add the following key-value pairs.

Key	Value
username	admin
password	DevOps@123

Click **Next**.

Key/value pairs Info

Key/value Plaintext

username	raji	Remove
password	rajumari	Remove

[+ Add row](#)

Encryption key Info

You can encrypt using the KMS key that Secrets Manager creates or a customer managed KMS key that you create.

aws/secretsmanager [Add new key](#)

[Cancel](#) [Next](#)

Step 5: Configure Secret

Enter:

Setting Value

Secret Name employee-db-secret

Description Database credentials for employee application

Click **Next**.

The screenshot shows the AWS Secrets Manager console during the 'Configure secret' step. The breadcrumb trail at the top reads 'AWS Secrets Manager > Secrets > Store a new secret'. On the left, a sidebar shows the 'Configure secret' step is selected. The main area is titled 'Configure secret' and contains two sections: 'Secret name and description' and 'Tags - optional'. In the 'Secret name and description' section, the 'Secret name' field is filled with 'employee-db-secret' and the 'Description' field is filled with 'Database credentials for employee application'. The 'Tags - optional' section shows 'No tags associated with the secret' and an 'Add' button.

Step 6: Configure Rotation

You will see the rotation settings.

For this lab:

Select

Disable automatic rotation

Reason: Automatic rotation requires a supported database (such as Amazon RDS) and an AWS Lambda function. Since this is a basic hands-on exercise, rotation is not required.

Click **Next**.

Rotation schedule [Info](#)

☒ Schedule expression builder ☐ Schedule expression

Time unit **Hours**

Hours

Window duration - optional

Enter the time in hours.

☒ Rotate immediately when the secret is stored. The next rotation will begin on your schedule.

Rotation function [Info](#)

Lambda rotation function [Info](#)

Choose a Lambda function that can rotate this secret.

[Create function](#)

[Cancel](#) [Previous](#) [Next](#)

Step 7: Review

Review all the settings.

Verify:

- Secret Name
- Secret Type
- KMS Key
- Rotation Status

Click

Store

Review

Secret type

Secret type
Other type of secret

Encryption key
aws/secretsmanager

Secret configuration

Secret name
employee-db-secret

Description
Database credentials for employee application

Tags
-

Step 8: Verify Secret

The new secret appears in the list.

Open it.

Observe:

- Secret ARN
- Secret Name
- Last Changed
- Encryption Key

Step 9: Retrieve Secret Value

Inside the secret:

Click

Retrieve secret value

Confirm.

You should see:

username : admin

password : DevOps@123

Step 10: Edit Secret

Click

Edit

Change the password to:

DevOps@456

Save the changes.

Step 11: View Secret Versions

Open the **Versions** tab.

Observe:

- Current Version

- Previous Version
- Version IDs

Step 12: Schedule Secret Deletion

After completing the lab:

1. Select the secret.
2. Click **Delete**.
3. Choose:

Recovery Window: 7 Days

4. Confirm.

Note: The secret is **scheduled for deletion** and can be restored within the recovery window.

In addition to using the AWS Management Console, we can also create and manage AWS SQS services using the AWS CLI by using below Commands

AWS CLI Commands (Optional)

List secrets:

❏ Bash

```
aws secretsmanager list-secrets
```

Create a secret:

❏ Bash

```
aws secretsmanager create-secret \  
  --name employee-db-secret \  
  --secret-string '{"username":"admin","password":"DevOps@123"}'
```

Retrieve the secret:

❏ Bash

```
aws secretsmanager get-secret-value \  
  --secret-id employee-db-secret
```

Delete the secret:

❏ Bash

```
aws secretsmanager delete-secret \  
  --secret-id employee-db-secret \  
  --recovery-window-in-days 7
```